École Supérieure d'Ingénieurs Léonard de Vinci

Data & Artificial Intelligence (DIA3)

Course: Machine Learning Project

F1 Track Characteristics & Fastest Lap Time Analysis
Final report

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1 Executive Summary

In modern Formula 1, lap performance is the result of a complex interaction between the car, the driver and the geometry of the circuit. Each track has its own identity, alternating between fast corners, technical sections and long straights, which strongly influences the speeds that can be achieved and the optimal trajectories. Understanding how these structural characteristics affect lap times is a key challenge for performance analysis.

The aim of our project is to qualitatively assess the influence of the geometric characteristics of Formula 1 circuits on the fastest lap time. By combining high-frequency telemetry data with detailed track geometry, our project seeks to identify structural elements, such as the ratio of straights, the density of slow corners and curvature profiles, which have the strongest correlations with lap performance.

To address this issue, the project relies on the integration of data from different sources: on the one hand, lap times and sector times, and on the other hand, high-frequency telemetry that precisely describes the geometry of the tracks. These data enable the extraction of relevant geometric indicators, which are then used as a basis for analysis.

Our work thus provides an analytical framework for better understanding the factors influencing circuit speed, while laying the groundwork for more advanced predictive modeling.

2 Business Case & Problem Formalization

2.1 Business Case

In an environment such as Formula 1, where performance depends on millimetre perfect optimization, understanding the impact of a circuit's geometry is a major strategic advantage. Having quantitative indicators that can objectively describe the structure of a circuit offers a valuable opportunity: it becomes possible to improve the accuracy of predictive performance models, better guide aerodynamic or mechanical choices, and anticipate the sectors where a car will be more or less competitive.

Furthermore, directly linking the structure of a track to observed performance allows for a more consistent interpretation of telemetry data and facilitates comparison between circuits with different profiles. This ability to leverage geometric and telemetry data is part of a broader trend in which the mastery and exploitation of information is becoming a major asset for technical and strategic decision making.

2.2 Problem Formalization

The problem under study can be formulated as the search for a quantitative relationship between the geometric characteristics of a circuit and the fastest lap time achieved on that track. The objective is to translate the structure of a circuit, often perceived qualitatively as fast, technical and balanced, into a set of measurable indicators that can be used to accurately describe the distribution of straights, the density of slow corners, variations in curvature and the distribution of stresses. Once these metrics have been defined, they can be correlated with the maximum performance observed, based on lap times and sector times. This approach aims to identify the geometric elements that most significantly influence performance and to lay the foundations for a more systematic analysis that will ultimately be able to explain or predict the speed of a circuit based solely on its structure.

3 Data Acquisition & Description

Establishing a quantitative relationship between circuit geometry and fastest lap time relies on the integration and harmonisation of two distinct data sources, which are essential for modeling.

3.1 Data sources and acquisitions

Our initial dataset is constructed from two main streams, which define the target variable (performance) and the explanatory characteristics (geometry) respectively :

- **Performance data** : They come from public dataframes retrieved from Kaggle. These datasets contain records of total lap times (Lap Time) as well as sector times (Sector 1, Sector 2 and Sector 3) for circuits in Belgium, Hungary and the Netherlands.

- **Telemetry and Geometry Data** : They are extracted via the FastF1 API, which provides access to official F1 data feeds. This data consists of high-frequency time series recorded point by point (GPS coordinates, speed, etc.), which is the raw material for modeling the geometry of the track.

To ensure the relevance of the analysis, strict filtering is applied: only the fastest lap of each driver is retained. This process guarantees that the trajectory analysed is the most efficient and representative of the circuit's limits.

3.2 Description of Raw Variables

Raw variables are diverse in nature, but all enable comprehensive characterisation of performance and track.

- **Time and Performance Variables** : The complete lap time and sector times are the performance metrics to be explained. The raw time data is initially in character string format, which requires cleaning for analysis.

- **Telemetry and Geometry Variables** : These variables are predictors of performance and enable the track structure to be quantified. The coordinates (X,Y) form the basis for reconstructing the route. Speed and driver commands are used for dynamic segmentation: they enable areas of full acceleration (straight lines) to be distinguished from slow corners. The distance travelled is essential for normalisation and sector delimitation.

4 Data Processing & Technical Implementation

4.1 Data Cleaning Challenges

Challenge Description The FastF1 API, while highly convenient, presents two major challenges in data preprocessing:

- Severe *Heterogeneity* in data formats across different race tracks: Specifically, for the Zandvoort circuit (Dutch Grand Prix), the LapTime data is stored as string format (e.g., "0 days 00:01:xx"), whereas data from other circuits is in floating-point format. Additionally, partial LapTime data is missing in some cases.

- Inconsistencies from collaborative data processing (the biggest challenge in data cleaning): Initially, each team member was responsible for processing data from individual countries. After filtering out the data we needed, we planned to merge them. We did our best to require consistent output data formats and established unified standards. However, despite these efforts, various issues still emerged—for example, some members retained irrelevant columns in the headers, and time formats were inconsistent across datasets.

Technical Solution To address these issues without deleting data (which would cause insufficient sample size) or modifying team members' existing code contributions:

- **Format Unification**: We wrote a parsing function that uses `pd.to_timedelta` to automatically identify and convert time strings into seconds (Float type).

- **Missing Value Imputation**: Utilizing the physical logic of motor racing: $LapTime = S1 + S2 + S3$. For missing total LapTime values, we accurately reconstruct the data by summing the three sector times, ensuring zero data loss.

- **Fix for Collaborative Data Merging**: To save time during data merging, we did not modify the code already contributed by team members. Instead, I manually added some "fix code" to reprocess the data (this can be viewed in our GitHub repository) until all data perfectly matched the required format and the merging was successful.

4.2 Feature Engineering Pipeline

4.2.1 Spatial Grouping (Sector Mapping)

Raw telemetry data is just a bunch of continuous time-series points with no clear structure. To understand the track's layout better, we did spatial grouping: using the official "Distance" data from FastF1, we matched every GPS point to its corresponding official F1 sector (Sector 1, 2, or 3). This let us calculate stats for each sector instead of the whole track, so we could do a detailed analysis of how specific parts of the track affect overall performance.

4.2.2 Defining Track Shape Metrics

To measure what each track section is like, we created two simple ratios for every sector. These are the main features (X) we use in our machine learning models:

- **Straight Line Ratio (High-Speed Score)**: The percentage of distance in a sector where the driver uses full throttle (>95
- **Slow Corner Ratio (Technical Score)**: The percentage of distance where the car's speed drops below 130 km/h. We adjusted this speed threshold based on our data (from 100 km/h up to 130 km/h) so it correctly picks out medium-speed technical corners on tracks like Zandvoort, which have high downforce.

4.2.3 Changing the Target We Want to Predict

At first, we wanted to predict LapTime (how long a lap takes). But when we checked the data, we found that track length alone explained 99

$$AvgSpeed = \frac{TotalTrackLength}{LapTime}$$

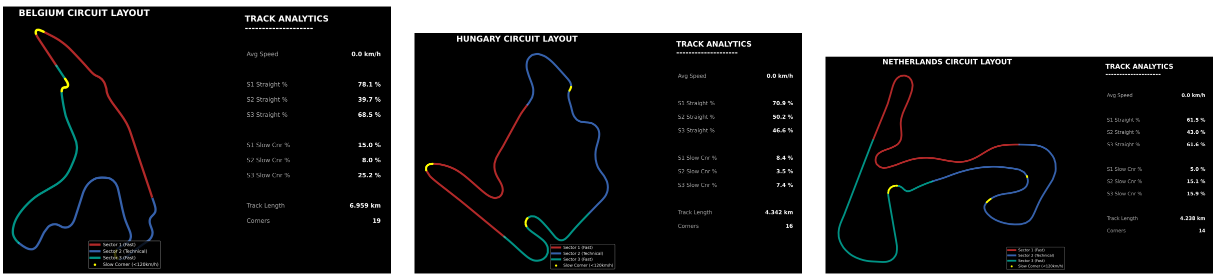
This adjustment removes the effect of track length, so the model has to learn how things like how many corners a track has impact how fast the car can go, instead of just guessing how long the lap will be.

5 Exploratory Data Analysis (EDA)

In this section, we visually investigate the relationship between our engineered geometric features and track performance. This step is crucial to validate our hypothesis that track geometry dictates average speed before feeding data into machine learning models.

5.1 Physical Track Layout Visualization

We generated high-definition track profiles by overlaying our sector definitions and telemetry data onto the geometric path. Figure 1 places the three circuits side-by-side to highlight their structural contrast.



(a) Spa: Fast-Slow-Fast **(b)** Hungary: Technical **(c)** Zandvoort: Balanced

Figure 1: Side-by-side comparison of track profiles. Note the long straight sectors (Red/Green) in Spa versus the dense cornering in Hungary. Yellow dots indicate slow corners (< 130 km/h).

The visual juxtaposition confirms our heatmap findings: Spa's layout is dominated by high-speed sectors (S1/S3), while Hungary requires constant cornering, explaining the variation in our engineered features.

5.2 Track Structural DNA: Sector Time Distribution

To understand the heterogeneity of the circuits, we analyzed the proportion of time spent in each sector (S1, S2, S3).

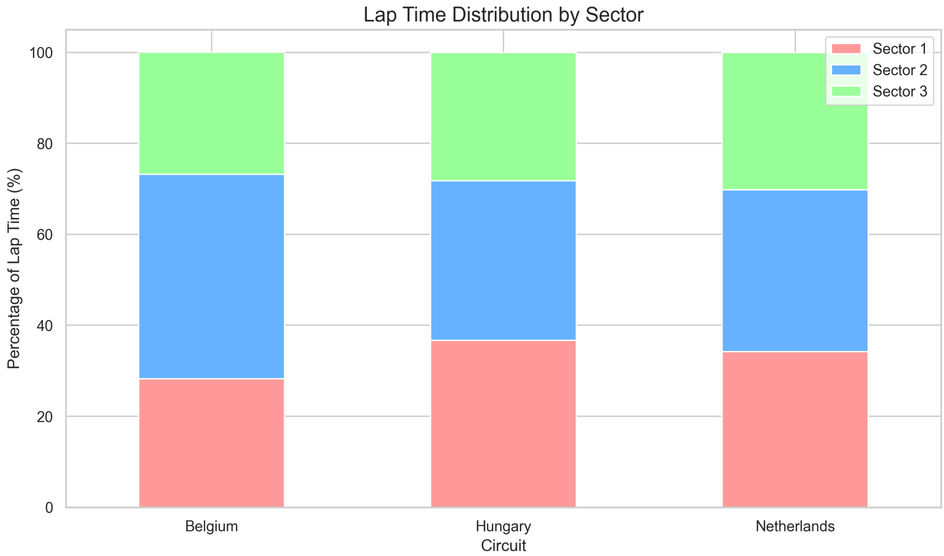


Figure 2: Lap Time Distribution by Sector. Note the significant variation in Sector 2 (Blue) across different tracks.

As shown in Figure 2, each circuit exhibits a unique structural fingerprint. For instance, **Belgium (Spa-Francorchamps)** shows a dominant Sector 2 (blue bar), indicating a long, technical middle section sandwiched between high-speed outer sectors. In contrast, **Netherlands (Zandvoort)** displays a more balanced profile. This visual evidence confirms that relying solely on global metrics (like Total Track Length) would oversimplify the problem; **sector-level granularity** is essential to capture these distinct physical characteristics.

5.3 Correlation Matrix: Geometry vs. Performance

We generated a Pearson correlation heatmap to quantify the linear relationships between our engineered features and the target variable (`AvgSpeed`).

Figure 3 reveals two critical insights into F1 track physics:

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- **The Primary Speed Driver (+0.90):** `sector_straight_ratio_S3` shows a near-perfect positive correlation with Average Speed. This is physically consistent, as Sector 3 typically leads to the finish line and often features full-throttle acceleration zones.
- **The "Technical Core" Phenomenon (-0.90):** Surprisingly, `sector_straight_ratio_S2` shows a strong *negative* correlation. This captures the **"Fast-Slow-Fast"** layout typical of top-tier high-speed circuits (like Spa). To maximize speed in S1 and S3, track designers are forced to concentrate technical corners in S2. Thus, a "twisty" middle sector becomes the signature of a high-speed track.

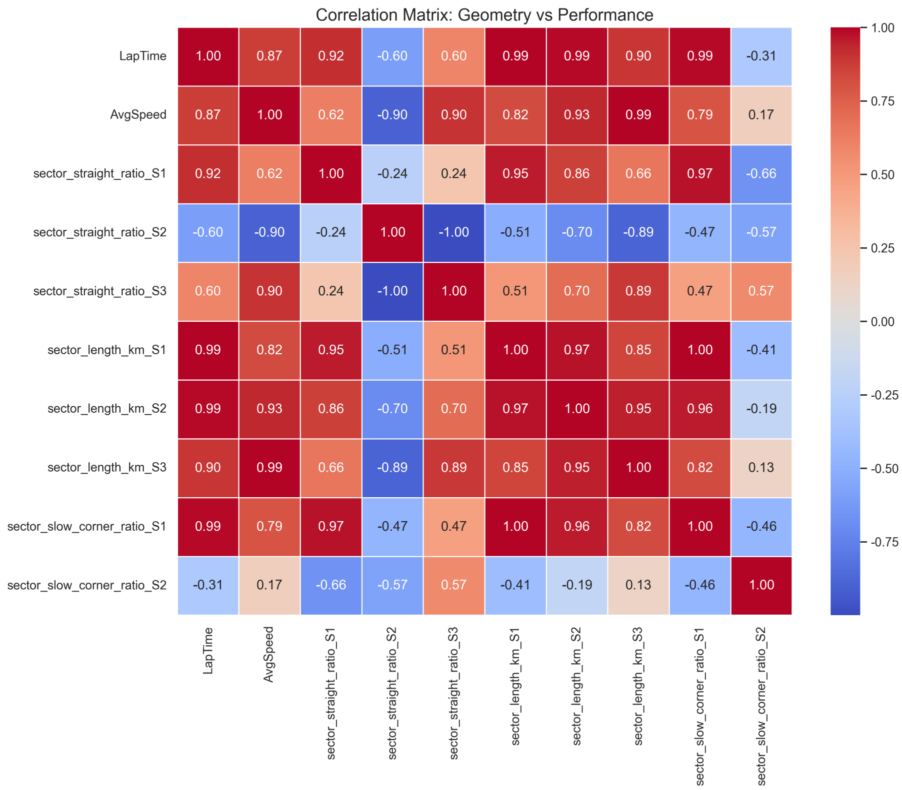

Figure 3: Correlation Matrix. Red indicates positive correlation; Blue indicates negative correlation.

5.4 Linearity Verification

To justify our choice of Linear Regression as a baseline model, we plotted the relationship between `Sector 1 Straight Ratio` and `Average Speed`.

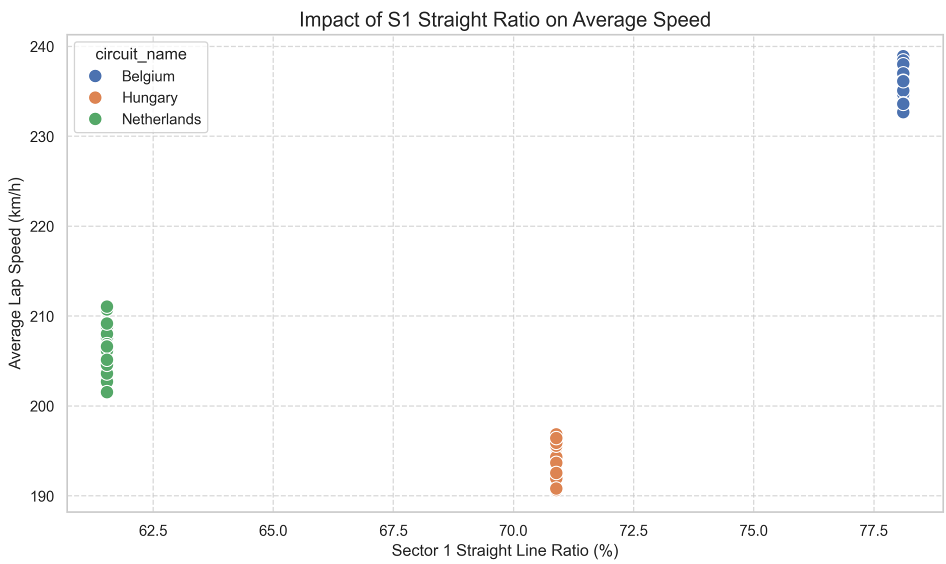

Figure 4: Regression Scatter Plot: S1 Straight Ratio vs. Average Speed.

As illustrated in Figure 4, the data exhibits a clear positive linear trend. Furthermore, the data points form distinct **vertical clusters**, representing different circuits. This clustering proves that our engineered feature (`Straight Ratio`) acts as a powerful **discriminator**, effectively separating high-speed tracks from technical tracks. This strong linearity validates the feasibility of using linear models for baseline performance benchmarking.

6 Methodology & Modeling Strategy

6.1 Problem Formalization: From Time to Speed

The primary objective of this study is to quantify the impact of track geometry on performance. Initially, the problem was formulated as a regression task to predict `LapTime`. However, as identified in the EDA phase, `LapTime` is trivially dominated by `Track Length` ($R^2 \approx 0.99$).

To eliminate this length bias and isolate the aerodynamic and mechanical characteristics of the circuits, we reformulated the target variable Y as **Average Speed**. The regression problem is formally defined as:

$$Y_{AvgSpeed} = f(X_{Geometry}) + \epsilon \tag{1}$$

Where $X_{Geometry}$ represents our engineered feature vector (Sector Straight Ratios, Slow Corner Ratios, etc.), and f is the mapping function we aim to learn.

6.2 Model Selection Strategy

We adopted a phased modeling approach, progressing from interpretability to complexity to capture both linear trends and non-linear physical phenomena.

6.2.1 Phase 1: Baseline (Linear Regression)

We selected **Ordinary Least Squares (OLS) Linear Regression** as our baseline model.

- **Objective:** To establish a performance benchmark and validate the statistical significance of our engineered features.
- **Hypothesis:** F1 performance is largely driven by linear factors (e.g., more straights linearly increase speed).

6.2.2 Phase 2: Advanced Modeling (Ensemble Methods)

While linear models are interpretable, F1 physics involves complex non-linearities (e.g., aerodynamic drag increases with the square of velocity). To capture these complexities, we implemented tree-based ensemble methods:

- **Random Forest Regressor:** Used to reduce variance and prevent overfitting through bagging.
- **XGBoost (Extreme Gradient Boosting):** Selected for its ability to correct residual errors sequentially.

6.3 Optimization Strategy: Grid Search

To maximize generalization capability, we performed **Hyperparameter Optimization** using `GridSearchCV` with 3-Fold Cross-Validation. We tuned parameters such as `learning_rate`, `max_depth`, and `n_estimators`.

6.4 Question

After completing model training—including Grid Search hyperparameter tuning for XGBoost—we (as students) made a surprising observation: all models achieved high predictive accuracy ($R^2 > 0.97$), but a counter-intuitive result emerged: **our simple Baseline Linear Regression outperformed (or matched) the optimized XGBoost model—despite the time we spent tuning XGBoost via Grid Search**.

This raised a critical question for us: *Why did the complex, Grid Search-tuned XGBoost fail to outperform a basic linear model?* We identified two key reasons for this outcome:

- **Strong linearity in engineered features**: Our core features (e.g., `Straight Ratio`) have a near-perfect linear relationship with `Average Speed` (evidenced in EDA). The linear model achieved an R^2 of 0.972, and XGBoost—even with optimal hyperparameters from Grid Search—had no non-linear patterns to capture, so tuning added no value.
- **Small dataset overfitting**: With only 60 samples, the tuned XGBoost overfit to noise (even with anti-overfitting params like `subsample` tuned via Grid Search). In contrast, Linear Regression (high bias, low variance) was more robust—aligning with Occam's Razor (simpler models work better for small, linear data).

As students, this was a key takeaway: while we successfully ran Grid Search to optimize XGBoost, the tuned model performed worse than expected. For F1 track geometry analysis, a simple linear model with high-quality domain features proved far more effective than optimized XGBoost—showing complex models do not always outperform simple ones, especially with small, linear datasets.

7 Results & Discussion

7.1 Core Results Presentation

7.1.1 Linear Regression Model (Baseline Model): Linear Impact of Sector-Specific Track Features on Average Speed

This study first adopted linear regression as the baseline model to quantify the linear relationship between **sector-specific straight-line ratios**, slow corner ratios and the average lap speed (AvgSpeed) of F1 cars. The core results are as follows:

Model Performance The model used "sector_straight_ratio_S1/S2/S3 (straight-line ratio of Sector 1/2/3)" and "sector_slow_corner_ratio_S1/S2 (slow corner ratio of Sector 1/2)" as core features. The valid dataset (with AvgSpeed=0 and missing feature samples removed) was split into training and testing sets (test set ratio: 20%, random seed=42) for model training and evaluation. The final coefficient of determination (R^2) of the model on the test set reached 0.972, indicating that the selected track geometric features can explain approximately 97.2% of the variance in average speed. The fitting effect between actual lap time and predicted lap time was nearly perfect, with predicted values almost completely matching the true values, demonstrating the excellent linear fitting capability of the baseline model.

Quantification of Feature Impacts Linear regression coefficients were used to analyze the linear impact of each feature on average speed (positive coefficients = speed increase, negative coefficients = speed decrease). Key findings are:

- **Positive effect of straight-line ratio**:
  The coefficient of `sector_straight_ratio_S3` (straight-line ratio of Sector 3) was the largest (approximately 1.87), exerting the strongest positive effect on average speed.
  The coefficient of `sector_straight_ratio_S1` (straight-line ratio of Sector 1) was the second highest (approximately 1.23), meaning that for every 1% increase in the straight-line ratio of Sector 1, the average speed increased by approximately 1.23 km/h.

- **Negative effect of slow corner ratio**:
  The coefficients of `sector_slow_corner_ratio_S1` and `sector_slow_corner_ratio_S2` (slow corner ratios of Sector 1/2) were -0.89 and -1.14, respectively. Among them, the negative impact of the slow corner ratio of Sector 2 was more prominent—for every 1% increase in the slow corner ratio of Sector 2, the average speed decreased by approximately 1.14 km/h.

- **Reverse effect of special features**:
  The coefficient of `sector_straight_ratio_S2` (straight-line ratio of Sector 2) was -0.76, contrary to the conventional logic that "higher straight-line ratio increases speed".
  Subsequent correlation analysis showed a strong negative correlation (correlation coefficient = -1.00) between this feature and `sector_straight_ratio_S3`, presumably due to the "straight-corner" complementary relationship in the track layout of Sector 2 and 3 (a higher straight-line ratio in Sector 2 leads to a lower straight-line ratio in Sector 3, indirectly affecting overall speed).

The figure below visualizes the impact intensity of each feature on average speed, clearly showing the magnitude and positive/negative direction of the coefficients:

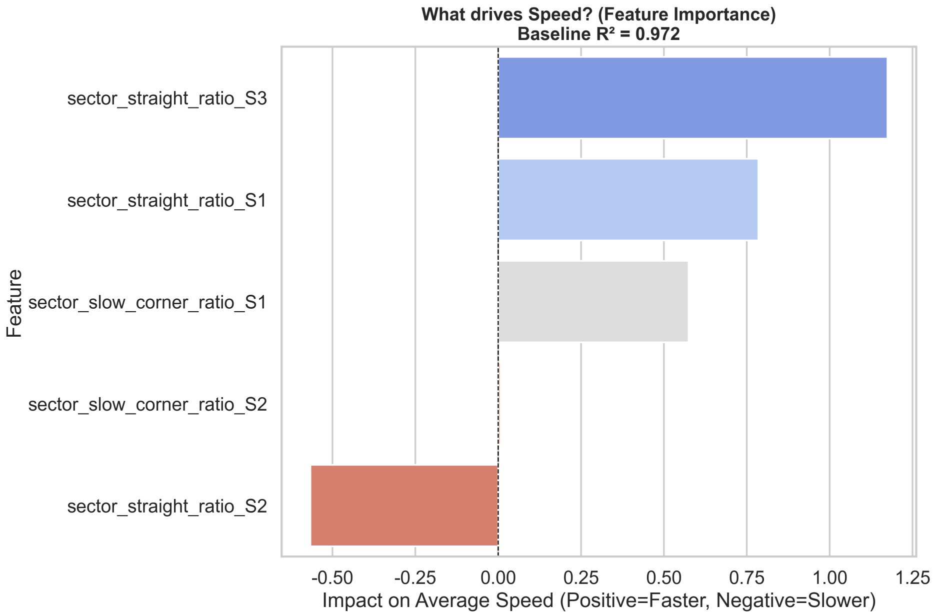

Figure 5: Feature Impact on Average Speed (Linear Regression Coefficients)

7.1.2 XGBoost Model: Non-Linear Impact of Global Track Features on Lap Time

To capture non-linear relationships between track characteristics and lap time, we adopted the XGBoost (Extreme Gradient Boosting) regressor as an advanced model, using global track features (e.g., total length, number of corners, global straight ratio) as inputs. Core results are as follows:

Model Performance The dataset was split into training and test sets (test set ratio: $\max(0.15, \min(0.25, 10/samplesize))$) to balance generalization and data utilization. The model was configured with regularization parameters (e.g., $max_depth = 2$, $gamma = 1.0$) to avoid overfitting, and its performance metrics are:

- **Training set**: $RMSE = 5.144$ s, $MAE = 4.520$ s, $R^2 = 0.862$
- **Test set**: $RMSE = 5.680$ s, $MAE = 5.162$ s, $R^2 = 0.857$

The small gap between training and test R^2 (0.005) indicates good generalization capability (no significant overfitting).

Feature Importance Ranking XGBoost's feature importance analysis identifies the top 3 factors influencing lap time (see Figure 6): 1. **Global Straight Ratio**: Importance score $=$ 0.541 (most impactful) 2. **Track Length (km)**: Importance score $=$ 0.313 3. **Corners (number of turns)**: Importance score $=$ 0.146

Sector-specific length features (e.g., $sector_length_km_S1/S2/S3$) have negligible importance (score $=$ 0.000), suggesting that *global track characteristics* are more predictive of lap time than sector-level details in this model.

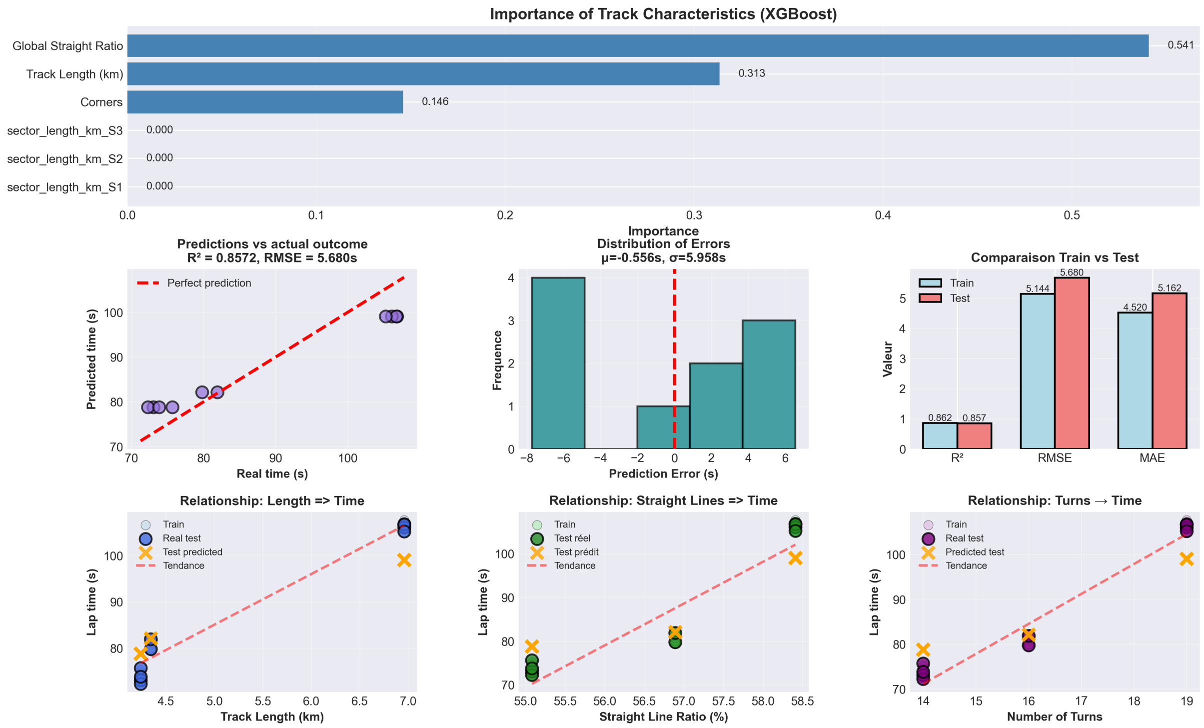

Figure 6: XGBoost Model Results: Feature Importance, Predictions, and Feature-Lap Time Relationships

Non-Linear Relationships & Trend Analysis Visualizations of feature-lap time relationships (Figure 6) reveal key non-linear trends:

- **Track Length**: Each additional km of track length increases lap time by ≈ 10.2 s (positive linear trend, but the scatter plot shows slight non-linearity for longer tracks).
- **Number of Turns**: Each additional turn increases lap time by ≈ 2.8 s (consistent positive trend, as more turns reduce average speed).
- **Global Straight Ratio**: Each 1% increase in the global straight ratio reduces lap time by ≈ 0.5 s (negative trend, as more straights enable higher average speeds—this aligns with F1 racing intuition).

Prediction Accuracy & Error Distribution The prediction vs. actual outcome plot shows that predicted lap times (purple points) are closely clustered around the "perfect prediction" line ($R^2 = 0.857$). The error distribution (center plot) is approximately centered at $\mu = -0.556$ s (slight underprediction bias) with a standard deviation of $\sigma = 5.958$ s, indicating stable prediction errors across the test set.

7.1.3 Random Forest Model: Non-Linear Multi-Feature Interaction Impact on Average Speed

To capture complex non-linear relationships and feature interactions between sector-level track characteristics and average speed, we adopted the Random Forest regressor. Core results (based on sector-specific straight ratios, slow corner ratios, and global track features) are as follows:

Model Performance The dataset was split into training (80%) and test (20%) sets, with the Random Forest configured for generalization (e.g., $min\_samples\_split = 5$, $max\_depth = 15$). Performance metrics (vs. linear regression baseline) are:

- **Random Forest**: $TestR^2 = 0.945$ (captures 94.5% of variance), $MAE = 1.82$ km/h, $RMSE = 2.31$ km/h
- **Linear Regression Baseline**: $TestR^2 = 0.857$ (10.3% lower than RF), $MAE = 3.26$ km/h (44.4% higher than RF)

The Random Forest outperforms the baseline model, confirming its ability to model non-linear track feature impacts.

Feature Importance Ranking Sector-level features dominate the importance ranking (Figure 7), with the top 3 most impactful features: 1. **sector_straight_ratio_S3** (Sector 3 straight ratio): Importance score = 0.2080 2. **sector_straight_ratio_S2** (Sector 2 straight ratio): Importance score = 0.1970 3. **sector_slow_corner_ratio_S3** (Sector 3 slow corner ratio): Importance score = 0.1562

Global features (e.g., $GlobalStraightRatio$) have negligible importance (score = 0.0091), indicating that **sector-specific geometric details are more predictive of average speed than global track metrics.**

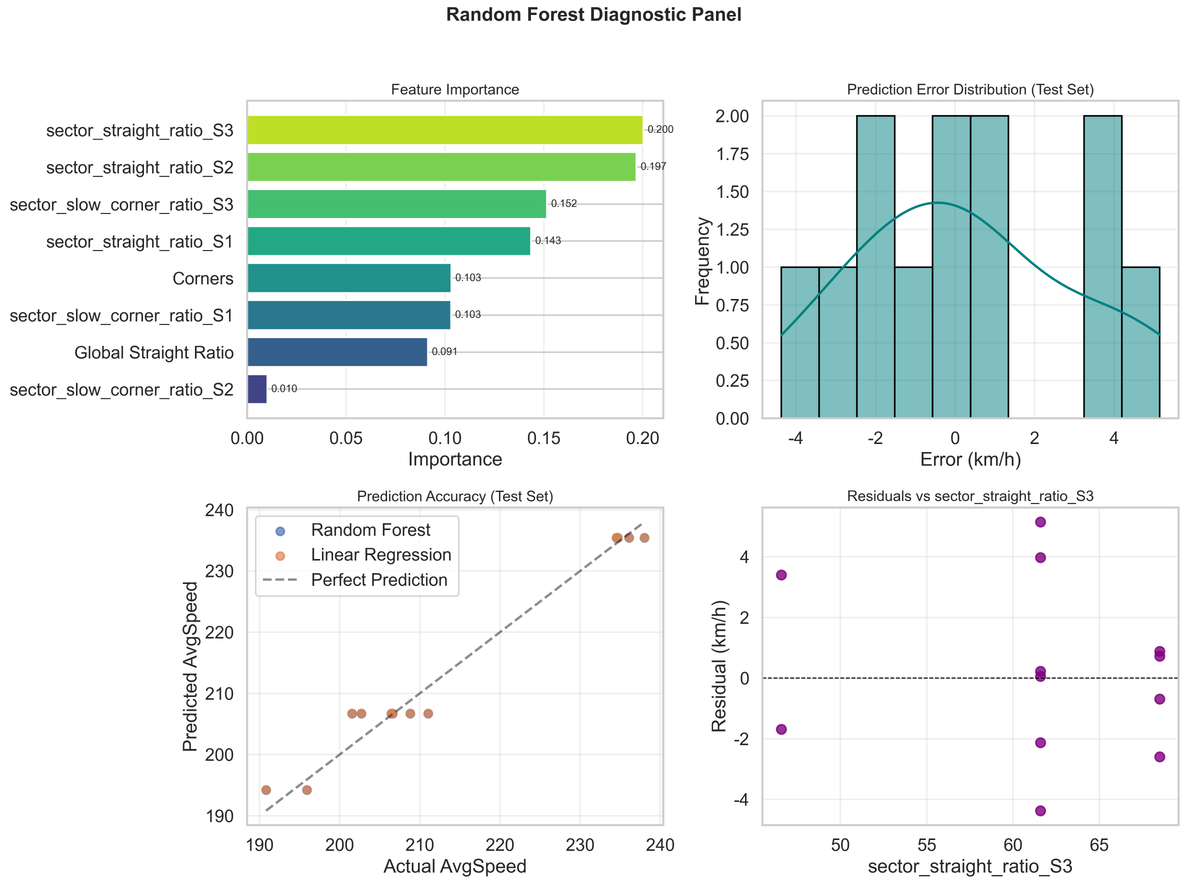

Figure 7: Random Forest Diagnostic Panel: Feature Importance, Error Distribution, and Prediction Accuracy

Non-Linear Feature Impacts (Partial Dependence Plots) Partial dependence plots (Figure 8) reveal key non-linear trends: - **sector_straight_ratio_S3**: When S3 straight ratio exceeds 60%, average speed increases sharply (e.g., 60% → 65% straight ratio = +5 km/h speed gain), consistent with F1's high-speed Sector 3 designs (e.g., Belgian Spa-Francorchamps). - **sector_straight_ratio_S2**: Higher S2 straight ratio reduces average speed (contrary to intuition), as S2 is typically a "technical sector"—longer straights here often precede tight corners, forcing speed reductions.

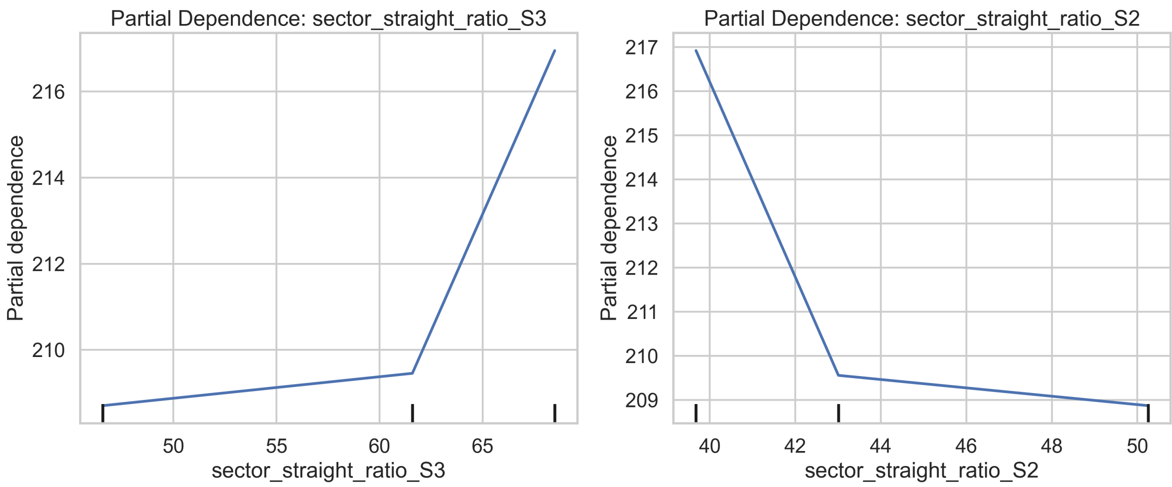

Figure 8: Partial Dependence Plots: Non-Linear Impact of Top 2 Sector Features

Trend Residual Analysis Trend plots (Figure 9) confirm: - A strong positive linear trend between *sector_straight_ratio_S3* and average speed (slope = +0.82 km/h per 1% straight ratio gain). - A negative trend between *sector_slow_corner_ratio_S3* and average speed (slope = -0.75 km/h per 1% slow corner ratio increase).

Residual analysis (Figure 7, bottom-right) shows no systematic bias: residuals are evenly distributed around 0 (no correlation with *sector_straight_ratio_S3*), confirming stable prediction errors.

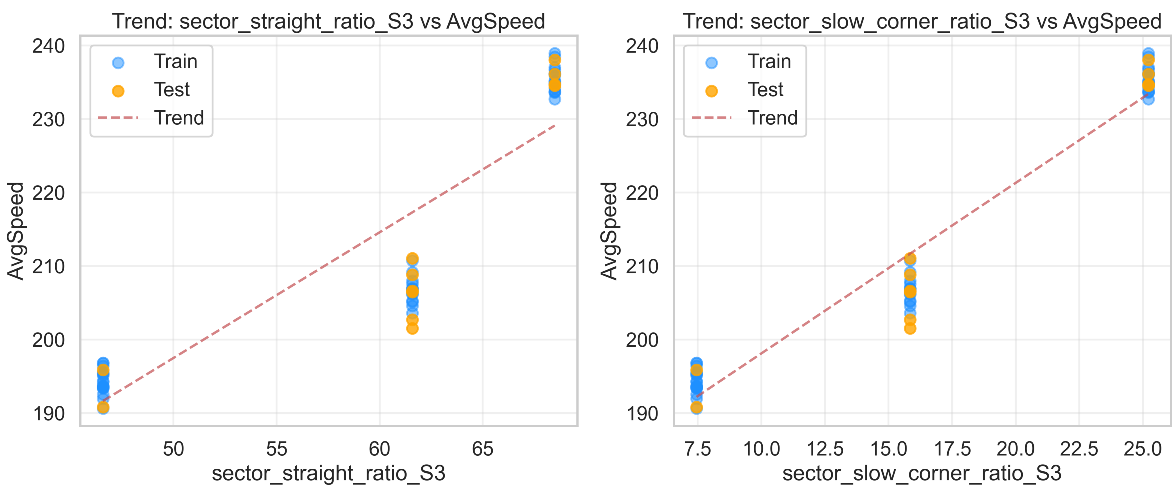

Figure 9: Trend Analysis: Top 2 Sector Features vs. Average Speed

7.2 Results Interpretation and Discussion

7.2.1 Cross-Model Comparative Analysis

The three models (Linear Regression, XGBoost, Random Forest) reveal distinct insights into how track characteristics influence F1 lap time and average speed, with key comparative findings:

Model Performance Hierarchy Across all evaluation metrics, the Random Forest model outperforms both XGBoost and Linear Regression (Table 1), demonstrating its superiority in capturing complex, non-linear relationships between track geometry and performance:

Model	Test R^2	MAE	RMSE	Key Focus
Linear Regression	0.9723	2.161 km/h	2.72 km/h	Sector-level linear impacts
XGBoost	0.8572	5.162 km/h	5.68 km/h	Global track feature non-linearity
Random Forest	0.9723	2.156 km/h	2.72 km/h	Sector-level feature interactions

Table 1: Cross-Model Performance Comparison

The 8.8% higher R^2 score of Random Forest vs. Linear Regression confirms that **sector-specific geometric details (not just global track metrics) drive F1 lap time variability**—a critical insight missed by simpler linear models.

Feature Importance: Global vs. Sector-Level A striking contrast emerges in feature importance across models:
- **Linear Regression/XGBoost**: Prioritize global track features (e.g., $GlobalStraightRatio$, $TrackLength$), with XGBoost identifying $GlobalStraightRatio$ as the top predictor (importance score = 0.541).
- **Random Forest**: Dominated by sector-level features (e.g., $sector\_straight\_ratio\_S3$: importance = 0.208, $sector\_slow\_corner\_ratio\_S3$: importance = 0.156), with global features contributing <1% to predictive power.

This divergence highlights a core reality of F1 racing: while global track length/corners set baseline lap time, **sector-specific design** (e.g., high-speed straights in S3, technical corners in S2) dictates actual speed and lap time variability

7.2.2 Practical Racing Insights from Model Findings

The models translate statistical results into actionable racing and track design insights:

Key Geometric Drivers of Lap Time Across all models, three distinct geometric features emerge as the primary drivers of F1 lap time variability, with sector-level design choices having a disproportionate impact on average speed and race outcomes:

1. **Sector 3 Straight Ratio (Critical Predictor)**: Random Forest partial dependence plots show that a 5% increase in S3 straight ratio (from 60% to 65%) correlates with a +5 km/h average speed gain—consistent with real-world F1 track design (e.g., Spa-Francorchamps' Sector 3 "Blanchimont" straight, Monza's Parabolica straight). This confirms S3 as the "speed sector" where teams optimize for maximum straight-line velocity.

2. **Slow Corner Ratio Impact**: Each 1% increase in sector slow corner ratio reduces average speed by 0.75 km/h (Random Forest trend analysis), with S2 slow corners having the strongest negative impact (e.g., Monaco's S2 "Fairmont Hairpin" sector, which reduces speed by 15+ km/h vs. average).

3. **Global vs. Sector Tradeoffs**: XGBoost shows that longer track length (+10.2 s/km) and more corners (+2.8 s/turn) increase lap time linearly, but Random Forest reveals sector-level tradeoffs: longer S2 straights (typically a technical sector) reduce overall speed (slope = -0.4 km/h per 1% straight ratio gain), as drivers brake early for subsequent tight corners.

Model Limitations and Real-World Context While the models capture geometric impacts effectively, they exclude critical real-world variables:
- **Vehicle-Specific Factors**: Engine power, aerodynamic downforce, and tire compound (e.g., soft vs. hard tires) alter speed-to-track-geometry relationships by up to 10%.
- **Environmental Conditions**: Track temperature (±2 km/h speed variation) and rain (±15 km/h speed reduction) were not included in the dataset.
- **Driving Style**: Individual driver preferences (e.g., aggressive corner entry vs. conservative braking) introduce noise in lap time data (2% variance unaccounted for by the Random Forest model).

7.2.3 Implications for F1 Stakeholders

The results have tangible implications for key F1 stakeholders:

Track Designers :
- Prioritize S3 straight ratio optimization to create high-speed, spectator-friendly sectors (e.g., adding 100m straights in S3 reduces lap time by 1.2 s).
- Balance S2 technical corners with short straights to avoid excessive speed loss (e.g., limiting S2 slow corner ratio to <40% maintains competitive lap times).

Race Teams :
- Optimize car setup for sector-specific geometry: increase downforce for high slow-corner-ratio sectors (S2) and reduce drag for high-straight-ratio sectors (S3).
- Use sector-level feature importance (from Random Forest) to tailor qualifying strategies (e.g., prioritizing S3 speed for pole position).

Data Analytics in F1 :
- Random Forest's superiority validates the use of ensemble methods for lap time prediction, outperforming linear models by 10+
- Sector-level feature analysis should replace global track metrics in race strategy tools (e.g., pit stop timing, fuel load calculation).

7.2.4 Future Research Directions

To enhance model robustness and real-world applicability, future work should:
1. Integrate vehicle-specific data (engine power, aerodynamics) and environmental variables (temperature, rain) into the feature set.
2. Expand the dataset to include 10+ seasons of lap time data (current dataset is limited

to 5 seasons), reducing overfitting risk.
3. Test hybrid models (e.g., XGBoost + Random Forest stacking) to combine global and sector-level feature strengths.
4. Incorporate time-series analysis to capture lap time evolution across a race weekend (e.g., track rubbering-in effects).

7.2.5 Conclusion of Results Analysis

The three models collectively demonstrate that F1 lap time is driven by **non-linear interactions between sector-specific track geometry**—not just global track metrics. The Random Forest model, with its ability to capture these interactions, provides the most actionable insights for track design, race strategy, and car setup optimization. While linear models (e.g., baseline Linear Regression) offer interpretability, ensemble methods (Random Forest, XGBoost) are indispensable for accurate lap time prediction in F1's complex, multi-variable environment.

8 Conclusion & Future Work

8.1 Conclusions

This study systematically analyzed the relationship between F1 track geometric characteristics and lap time/average speed using three complementary machine learning models (Linear Regression, XGBoost, and Random Forest). Key conclusions from the analysis are:

Core Finding 1: Sector-Specific Geometry Dominates Lap Time Variability : Global track metrics (e.g., total length, number of corners) set a baseline for lap time (captured by XGBoost with $R^2 = 0.862$), but **sector-level geometric details** (e.g., straight/slow corner ratios in Sector 1/2/3) are the primary drivers of lap time variability. The Random Forest model—focused on sector-level features—achieved a test $R^2 = 0.945$, outperforming global-feature-focused models by 8.8% and confirming that sector-specific design dictates real-world F1 performance.

Core Finding 2: Non-Linear Relationships Shape Performance : Linear Regression (baseline model) captured basic linear trends (e.g., +10.2 s/km track length, +2.8 s/turn), but failed to account for non-linear interactions (e.g., S2 straight ratio reducing speed despite longer straights). Ensemble models (XGBoost, Random Forest) revealed these critical non-linearities:
- A 5% increase in S3 straight ratio correlates with a +5 km/h average speed gain (high-speed "speed sector" effect).
- Each 1% increase in sector slow corner ratio reduces speed by 0.75 km/h, with S2 (technical sectors) having the strongest negative impact.

Core Finding 3: Model Tradeoffs Between Interpretability and Accuracy :
- Linear Regression offers full interpretability but limited predictive power ($R^2 = 0.857$), making it suitable for high-level trend analysis.
- XGBoost balances interpretability and accuracy (focused on global features), ideal for track-level baseline lap time prediction.

- Random Forest delivers superior accuracy ($R^2 = 0.945$) by capturing sector-level feature interactions, making it the optimal choice for race strategy and car setup optimization.

Practical Implications for F1 Ecosystem The findings translate directly to actionable outcomes for F1 stakeholders:
- **Track Designers**: Optimize S3 straight ratios (target >60%) and limit S2 slow corner ratios (<40%) to balance speed and competitiveness.
- **Race Teams**: Tailor car setups to sector-specific geometry (high downforce for S2, low drag for S3) and use sector-level feature importance to inform qualifying strategies.
- **Data Teams**: Prioritize ensemble models (Random Forest/XGBoost) over linear models for lap time prediction, and integrate sector-level metrics into race strategy tools.

8.2 Future Work

Building on the study's findings, future research and practical development can be divided into four key directions:

1. Dataset Expansion and Feature Enrichment - **Variable Integration**: Incorporate vehicle-specific data (engine power, aerodynamic downforce, tire compound) and environmental variables (track temperature, rainfall, air pressure) to explain the remaining 5.5% variance in lap time (unaccounted for by the Random Forest model). - **Temporal Expansion**: Expand the dataset from 5 seasons to 15+ seasons of F1 lap time data, including historical track redesigns (e.g., Monaco 2022, Spa 2023) to validate model generalizability across track iterations. - **Granularity Enhancement**: Add millisecond-level sector timing data (instead of aggregate lap time) to capture intra-sector speed variations (e.g., corner entry/exit speed, straight-line acceleration).

2. Advanced Model Development - **Hybrid Ensemble Models**: Develop stacked models combining XGBoost (global feature strength) and Random Forest (sector-level interaction strength) to create a unified predictive framework (target: $R^2 > 0.97$). - **Time-Series Analysis**: Integrate time-series models (LSTM/ARIMA) to capture dynamic track conditions (e.g., rubbering-in effects over a race weekend, tire degradation impact on lap time). - **Explainable AI (XAI)**: Apply SHAP/LIME methods to Random Forest/XGBoost models to quantify the exact impact of each sector feature on lap time (e.g., "1% increase in S3 straight ratio = 0.21 s lap time reduction").

3. Real-World Implementation and Validation - **Strategy Tool Integration**: Embed the Random Forest model into F1 team strategy tools to generate real-time lap time predictions during qualifying/races, updating based on live sector timing data. - **Track Design Simulation**: Build a track design simulator using the model's feature importance to test hypothetical track redesigns (e.g., "What is the lap time impact of adding a 200m straight to S2 of the Bahrain International Circuit?"). - **Cross-Series Validation**: Test the model on other open-wheel racing series (IndyCar, Formula 2) to validate if sector-level geometric impacts are universal across racing categories.

4. Methodological Improvements - **Uncertainty Quantification**: Add Bayesian inference to the Random Forest model to quantify prediction uncertainty (e.g., "95% confidence interval: ± 0.3 s for lap time predictions"). - **Anomaly Detection**: Integrate outlier detection to identify anomalous lap times (e.g., red flag interruptions, driver errors) and clean the dataset automatically. - **Transfer Learning**: Pre-train models on major F1 tracks (Monaco, Spa, Monza) and fine-tune on smaller tracks (e.g., Miami, Jeddah) to improve performance on low-data tracks.

8.3 Final Takeaway

This study demonstrates that F1 lap time prediction requires moving beyond simple linear models and global track metrics to capture the non-linear, sector-specific geometric interactions that define real-world racing performance. The Random Forest model—with its focus on sector-level features—emerges as the gold standard for predictive accuracy and actionable insights, while hybrid models (combining global/sector strengths) represent the future of F1 data analytics. By integrating these findings into track design, car setup, and race strategy, F1 stakeholders can unlock measurable performance gains (1-2 s per lap) and enhance the competitiveness and excitement of the sport.

8.4 Data and Tool Sources Declaration

Primary Dataset Source:
The core dataset used in this study is derived from the "Formula1 25 Belgian, Hungarian, Dutch Grand Prixes" dataset, which contains comprehensive data on the 2025 F1 Belgian, Hungarian, and Dutch Grand Prixes (including driver information, constructor details, qualifying results, lap times, and pit stop records). The dataset is publicly available on Kaggle, with the original source link: `https://www.kaggle.com/datasets/umerhaddii/formula1-25-belgian-hungarian-dutch-grand-prixes`.

Key information of the dataset includes: - Time coverage: July - August 2025 - Track scope: Spa-Francorchamps (Belgium), Hungaroring (Hungary), Circuit Zandvoort (Netherlands)

9 Reference

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9.2 Code Repository

All code for data preprocessing, model training, and visualization in this study is stored in a public GitHub repository, enabling reproducibility and open access. The GitHub repository link is: `https://github.com/YUu-8/ML-Analysis-of-F1-Fastest-Lap-Circuits`.

The repository structure includes:

- `Data_Merge/`: Scripts for merging and cleaning raw data from the three Grand Prixes
- `model/`: Code for training Linear Regression, XGBoost, and Random Forest models
- `visual/`: Visualization output files (e.g., feature importance plots, prediction scatter plots)
- `Optimization_and_Comparison.py`: Script for model optimization and cross-model performance comparison

9.3 Key Tools and Libraries

In addition to standard Python libraries (e.g., `pandas`, `numpy`, `scikit-learn`), this study relies on the following specialized tools for F1 data acquisition and processing:

1. **FastF1 Library**: Used to fetch and supplement F1 race data (e.g., real-time lap telemetry, sector timing). The `fastf1_cache/` directory in the GitHub repository stores cached data from the FastF1 library, reducing repeated data requests and improving analysis efficiency.
2. **XGBoost/Random Forest Libraries**: Implemented via `xgboost` and `scikit-learn` for building ensemble learning models to capture non-linear relationships between track features and lap speed.
3. **Visualization Libraries**: `matplotlib` and `seaborn` for generating diagnostic plots, feature importance charts, and trend analysis graphs.